Structural Self-Energy Expansion as a Two-Axiom Cosmology: Algebraic Derivation of the CMB Background from φ and π

SSEE — Paper 4

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Abstract

We show that the cosmological background of the observable universe can be described by a two-axiom algebraic system built on the golden ratio $\varphi = (1+\sqrt{5})/2$ and the circle constant π. A third structural constant, $\Omega = \varphi + \pi$, is a definition rather than an independent axiom; no free parameter is introduced at the background level. Phenomenological extensions (notably the neutrino mass sum Σm_ν, formerly classified as Type P) were tracked as open problem OP-14 in `OPEN_PROBLEMS.md`, resolved on 2026-06-04. From these two axioms — and, for the Hubble rate, the measured SH0ES rate as the single dimensional input — we derive, without fitting:

(i) the pure number $3(\varphi+\pi)^2 = 67.962137$, which read in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ lies 1.1σ from Planck 2018. It is the comparison *target* of the Hubble sector, not an input: the model's rate is the output of the SH0ES cascade in (iv), and it lands on this number (Paper 10), where $M_v = 3\Omega$ is the maximal structural invariant of the dictionary, reached by 24 distinct additive routes under the no-self-sum construction law (Appendix A); this multiplicity is consistent with, but does not by itself prove, the absence of post-hoc tuning;

(ii) the physical baryon density $\Omega_b h^2 = (\pi-\varphi)/(3\Omega^2) = (\pi-\varphi)/[3(\varphi+\pi)^2] = 0.022418$ (0.32σ from Planck 2018);

(iii) the primordial spectral index $n_s = 1-\varphi^{-7} = 0.965558$ (0.16σ from Planck 2018), where the exponent 7 is fixed by Postulate I of Paper 1 ($N_* = 2\varphi^n$ e-folds falls in the inflation window $[50, 60]$ only for $n = 7$);

(iv) the global Hubble rate of the screening cascade of Papers 9–10, with Riess et al. [2022] as the input: $H_0^{\mathrm{glob}} = H_0^{\mathrm{SH0ES}}(1 - f_{\mathrm{screen}}) = 67.962142 \pm 0.968$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ with the complete screening fraction (with the IR term alone, $H_0^{\mathrm{glob,IR}} = 68.13$, 0.17σ from $3(\varphi+\pi)^2$); the earlier CMB-mapped register (via $H_0^{\mathrm{MIRA}} = 67.037$) is superseded; see Section 6).

These background predictions are number-to-number comparisons with zero adjustable parameters in their respective sectors. The same two axioms fix the cosmological background — including the

dark-energy parameters $w_0 = -0.839950$, $w_a = -0.669975$ derived in the companion Papers 1–3. The static relation gives $\Omega_c h^2 = K_{\rm AL}\,\Omega_b h^2 = 0.1238$ ($+3.1\sigma$); with the primordial tilt $n_s = 1-\varphi^{-7}$ as a factor it becomes $\Omega_c h^2 = K_{\rm AL}\,\Omega_b h^2\, n_s = 0.119514$ (-0.40σ from Planck), within 1σ of the observed value. A falsifiable prediction $r = \varphi^{-10} = 0.008131$ is presented for the LiteBIRD satellite mission ($\sim$2032).

Contents

1 Introduction **4**

2 The SSEE Two-Axiom System **4**
2.1 Axioms 4
2.2 Definition: $\Omega = \varphi + \pi$ 4
2.3 The Maximal Structural Invariant M_v 5

3 Dimensional Analysis: Honest Statement of Scope **5**
3.1 Internal Consistency of the Cosmological Predictions 6

4 Algebraic Derivation of Cosmological Parameters **6**
4.1 Hubble Constant H_0 6
4.2 Physical Baryon Density Ω_b 7
4.3 Total Matter Density Ω_m 8
4.4 Primordial Spectral Index 8
4.5 Linear Collapse Threshold δ_c 9
4.6 Big Bang Nucleosynthesis: Helium-4 Fraction 10
4.7 Dark-Energy Equation of State 10
4.8 Open Parameters 10

5 CMB Power Spectrum Prediction **11**

6 Note on the Hubble Tension **11**

7 Falsifiability Programme **11**
7.1 Tensor-to-Scalar Ratio 11
7.2 Baryon-to-Photon Ratio 12
7.3 Neutrino Mass Sum 12

8 Discussion and Response to Peer Review **13**

9 Conclusion **14**

A The Additive Routes to M_v **15**

1 Introduction

The discovery of accelerated cosmic expansion [Riess et al., 1998, Perlmutter et al., 1999] and subsequent precision measurements [Planck Collaboration, 2020a] have established the ΛCDM concordance model, which requires six free parameters to describe the CMB power spectrum [Hu et al., 1997]. Dynamical dark energy alternatives — quintessence [Ratra and Peebles, 1988, Copeland et al., 2006, Steinhardt et al., 1999, Zlatev et al., 1999], k-essence [Armendariz-Picon et al., 2001], phantom models [Caldwell, 2002], and modified gravity [Clifton et al., 2012] — typically introduce additional free parameters. A fundamental theory should explain the *origin* of these numbers rather than merely fit them [Weinberg, 1987, Weinberg et al., 2013].

The SSEE framework (Structural Self-Energy Expansion, [Almeida Vallejo, 2026a]) derives the dark-energy equation-of-state parameters $(w_0, w_a) = (-0.839950, -0.669975)$ in the Chevallier–Polarski–Linder parametrization [Chevallier and Polarski, 2001, Linder, 2003] algebraically from φ and π, with zero free parameters in that sector [Almeida Vallejo, 2026b,c,d]. Here we extend this programme to the entire cosmological background. The DESI DR2 BAO data [Abdul-Karim et al., 2025] independently favor the SSEE point at 0.24σ (Pantheon+; up to 1.8σ across the other SN compilations) in the w_0–w_a plane [Almeida Vallejo, 2026c].

Peer-review notice. An internal audit identified four vulnerabilities in a prior draft of this paper: (A) use of dimensional quantities without unit justification; (B-1) treating $\Omega = \varphi + \pi$ as an axiom rather than a definition; (B-2) post-hoc selection of the exponent in $n_s = 1 - \varphi^{-7}$; (B-3) unexplained factor of 100 in the baryon density. This revised manuscript addresses each point explicitly.

2 The SSEE Two-Axiom System

2.1 Axioms

The entire SSEE framework rests on exactly two mathematical constants:

$$\varphi = \frac{1+\sqrt{5}}{2} = 1.618033988\ldots \quad \text{(golden ratio)} \tag{1}$$

$$\pi = 3.141592653\ldots \quad \text{(circle constant)} \tag{2}$$

φ is algebraic (a root of $x^2 - x - 1 = 0$) and π is transcendental; both are defined independently of any physical measurement.

2.2 Definition: $\Omega = \varphi + \pi$

The SSEE structural constant Ω is **not** an independent axiom. It is defined as the sum of the two axioms:

$$\Omega \equiv \varphi + \pi = 4.759626642\ldots \tag{3}$$

This single expression generates the entire hierarchy of SSEE constants (Table 1) through algebraic operations alone.

2.3 The Maximal Structural Invariant M_v

The value

$$M_v \equiv \varphi + \pi + K_v = 3\,\Omega = 14.278879927\ldots \tag{4}$$

is the *maximal structural invariant* of the constant dictionary. Under the additive construction law—each constant is a sum of distinct previously defined parents, with no constant added to itself—it is the closure value at which the dictionary terminates, coinciding through $M_v = 3\Omega$ with the copy-law ceiling.

A direct enumeration under this law shows that M_v is reached by *24* distinct additive routes built from the named constants: 8 using two constants, 13 using three, and 3 using four (Appendix A). Representative routes, in neutral notation, are

$$\Omega + K_v, \quad \beta + \mathrm{KAL} + P_{sc}, \quad P_{sc} + \Omega + \pi, \quad \varphi + \pi + K_v, \tag{5}$$

each equal to M_v to 16 significant digits. This multiplicity is a consequence of M_v being the maximal invariant of the two-dimensional lattice spanned by φ and π; it is consistent with, but does not by itself prove, the absence of post-hoc tuning.

Table 1: SSEE structural constants. All entries are algebraic in φ and π only; no empirical input is used.

Name	Formula in φ, π	Simplified	Value
Ω	$\varphi + \pi$	—	4.759627
β	$(\varphi + \pi)/2$	$\Omega/2$	2.379813
AURA	$(3\varphi + \pi)/2$	—	3.997847
MIRA	$(3\varphi + \pi)/4$	AURA/2	1.998924
KAL	$(\varphi + 3\pi)/2$	—	5.521406
P_{sc}	$2\varphi + \pi$	$\Omega + \varphi$	6.377661
$\Omega + \pi$	$\varphi + 2\pi$	—	7.901219
K_v	$2\varphi + 2\pi$	$\varphi + \pi + \Omega$	9.519253
$\pi + \mathrm{KAL}$	$(\varphi + 5\pi)/2$	—	8.662999
$\varphi + \pi + \mathrm{KAL}$	$(3\varphi + 5\pi)/2$	—	10.281033
$2\Omega + \varphi$	$3\varphi + 2\pi$	—	11.137287
$\Omega_b h^2$ (BBN)	$(\pi - \varphi)/[3(\varphi + \pi)^2]$	—	0.022418
M_v	$3(\varphi + \pi)$	$\varphi + \pi + K_v$	14.278880

3 Dimensional Analysis: Honest Statement of Scope

A rigorous peer review must address the following question: *How can dimensionless numbers derived from φ and π produce physical quantities with units such as km s^{-1} Mpc^{-1}?*

Answer. They cannot directly; consequently, SSEE treats these relations only as dimensionless numerical correspondences rather than literal derivations of physical units. What SSEE claims is strictly the following:

Scope of SSEE Predictions

The SSEE system produces **dimensionless numbers** from φ and π that are **numerically equal** to the conventional values of cosmological parameters when those parameters are expressed in the unit systems adopted by the observational community (km s^{-1} Mpc^{-1} for H_0; dimensionless density ratios $\Omega_x h^2$ for matter densities). The claim is not "units are derived"; the claim is "the numerical values are predicted".

3.1 Internal Consistency of the Cosmological Predictions

A single dimensionless coincidence could be dismissed as numerology. What is harder to dismiss is a *system* of mutually consistent cosmological predictions — H_0, Ω_b, n_s, and the Hubble-tension scale — derived from the *same two constants* with *zero adjustments*, assessed against the closed-dictionary look-elsewhere analysis of Paper 1.

Table 2 collects these predictions.

Table 2: Cross-domain predictions of the SSEE two-axiom system. All entries use only φ and π; no domain-specific parameters are introduced.

Domain	SSEE formula	SSEE value	Observed	Deviation
Cosmology				
H_0 target (pure number)	$3(\varphi+\pi)^2$	67.962137	67.36 ± 0.54	1.1σ
$H_0^{\text{glob,IR}}$ [km/s/Mpc]	$H_0^{\text{SH0ES}}(1-f_{\text{scr}}^{\text{IR}})$	68.13	$3(\varphi+\pi)^2$	0.17σ
Ω_b	$\frac{10^4(\pi-\varphi)}{27(\varphi+\pi)^6}$	0.0485	0.0493 (ω_b/h^2)	composite*
n_s (inflation)	$1-\varphi^{-7}$	0.965558	0.9649 ± 0.0042	0.16σ

P IR-cascade value ($M \to \infty$ regime, not canonical): $f_{\text{scr}}^{\text{IR}}$ applied to the measured $H_0^{\text{SH0ES}} = 73.04$ returns 68.13, i.e. 0.17σ from $3(\varphi+\pi)^2 = 67.962137$; the full IR+UV cascade gives the canonical $H_0^{\text{glob}} = 67.96214$ (Paper 10). The earlier CMB-mapped register ($H_0^{\text{MIRA}} = 67.037$) is superseded.
Consistent with Paper 1 / Paper 9 policy (single target shared by every sector; the measured SH0ES rate is the input).
* $\Omega_b = \omega_b/h^2$ is composite; its inputs are compared separately. It is -1.6% from Planck's, the H_0 offset.

These cosmological ratios are evaluated against a fixed, closed algebraic dictionary built from $\{\varphi, \pi\}$; a quantitative look-elsewhere analysis of that dictionary is given in Paper 1 and the consolidated dark-energy document, which is the proper basis for assessing global significance rather than an informal product of individual deviations.

4 Algebraic Derivation of Cosmological Parameters

4.1 Hubble Constant H_0

From Section 2.3, the maximal structural invariant is $M_v = 3(\varphi+\pi)$. The Hubble target is the product of this convergent sum and the base unit $\Omega = \varphi + \pi$:

$$\boxed{M_v \times \Omega = 3(\varphi+\pi)^2 = 67.962137} \tag{6}$$

Dimensional note. The number 67.962137 is pure and carries no units, so it cannot be the input of a dimensionful cascade. The model's rate is the output of the SH0ES cascade, $H_0^{\text{glob}} = H_{\text{SH0ES}}(1 - f_{\text{screen}}) = 67.962142$ km s^{-1} Mpc^{-1} (Papers 9–10), and the pure number is the target that output is compared against. Why the measured cascade lands on it would require deriving H_0 from first principles — a quantum-gravity connection between the Planck length and the Megaparsec that no cosmological model has achieved — and is left open.

Global rate vs. observational posterior. The pure number $3(\varphi+\pi)^2 = 67.962137$ is a geometric target, not an input. Baryon acoustic oscillations [Eisenstein et al., 2005]

provide a standard ruler; independent Bayesian inference against DESI DR2 BAO data [Abdul-Karim et al., 2025] yields $H_0^{\rm MCMC} = 67.82 \pm 0.41$ km s^{-1} Mpc^{-1} (Paper 2, under the global rate of the screening cascade as prior, $H_0^{\rm glob} = H_{\rm SH0ES}(1 - f_{\rm screen}) = 67.962 \pm 0.968$, Papers 9–10). With the total gravitating matter density $\Omega_m = \omega_m/h^2 = 0.308881$ in the background geometry, the BAO-corrected posterior is consistent with the global rate $H_0^{\rm glob}$ at 0.34σ, and the SSEE sound horizon matches ΛCDM ($r_d^{\rm SSEE} = 147.17$ Mpc vs. $r_d^{\Lambda\rm CDM} = 147.10$ Mpc, CAMB): DESI DR2 is consistent with the global rate without singling it out. The $\sim$0.8 km s^{-1} Mpc^{-1} downshift reported in earlier drafts was an artefact of using the cold sector $1 + w_0 = 0.160050$ as the background density, now corrected. $H_0^{\rm glob}$ is thus the SSEE background rate obtained from the measured SH0ES input; $H_0^{\rm MCMC}$ is the same quantity after the BAO correction is applied, and the two agree to 0.34σ.

4.2 Physical Baryon Density Ω_b

The SSEE algebraic structure yields the physical baryon density directly from the CP asymmetry $(\pi - \varphi)$ normalized to the pure algebraic combination $3\Omega^2 = 3(\varphi + \pi)^2$, whose value read in km s^{-1} Mpc^{-1} coincides with the global rate $H_0^{\rm glob}$ of the SH0ES cascade:

$$\Omega_b h^2 = \frac{\pi - \varphi}{3\Omega^2} = \frac{\pi - \varphi}{3(\varphi + \pi)^2} = 0.022418 \tag{7}$$

$$\Omega_b = \frac{\Omega_b h^2}{h^2} = \frac{10^4(\pi - \varphi)}{27(\varphi + \pi)^6} = 0.04854 \tag{8}$$

Planck 2018: $\Omega_b h^2 = 0.02237 \pm 0.00015$; deviation $= 0.32\sigma$.

Status: an algebraic ansatz, with no baryogenesis mechanism behind it. Eq. (7) is an algebraic ansatz. Earlier drafts motivated its *form* with the Sakharov conditions [Sakharov, 1967] — electroweak sphalerons, a CP phase $\delta_{\rm CP} = (\pi - \varphi)/\Omega$, and departure from equilibrium through a late reheating — read as $\eta_B \propto \delta_{\rm CP}/(\text{expansion rate})$, with $(\pi - \varphi)$ as numerator and $3\Omega^2$ as denominator. That mechanism is *withdrawn*: the reheating it requires, $T_{\rm rh} \approx 10^{-4}$ GeV, lies below the BBN lower bound $T_{\rm rh} \gtrsim 4 \times 10^{-3}$ GeV [de Salas et al., 2015] and six orders of magnitude below the temperature at which sphalerons are active, so the sphalerons it invokes would never have run. The formula does not depend on it and does not move. Two caveats on the form remain. First, $3\Omega^2$ enters as the dimensionless value 67.962137, which coincides numerically with the global rate $H_0^{\rm glob}$ read in km s^{-1} Mpc^{-1}; this is a Type-P numerical coincidence, not a dimensional identity (in Postulate D of Paper 1 the dimensional input is the measured SH0ES rate). Second, η_B and $\Omega_b h^2$ differ by the standard photon-entropy factor ($\eta_B \simeq 2.74 \times 10^{-8}\, \Omega_b h^2$), so the proportionality fixes the algebraic *form* only, not the absolute normalisation. The formula was chosen among 7 φ, π expressions compared with Planck; it is the only one within 1σ, the next being at 3.2σ (the scan is reproduced by `src/p04_toe/barrido_omega_b.py`). This is a selection statement, not a dynamical derivation. A derivation of ω_b from first principles is the residual part of OP-1. $\Omega_b^{\rm SSEE} = 0.0485$ is -1.6% from the Planck ΛCDM value 0.0493 ($= \omega_b/h^2$ with Planck's own inputs). $\Omega_b = \omega_b/h^2$ is composite: its two inputs are compared separately (ω_b at 0.32σ, h through H_0), so this difference is the H_0 offset, not an independent tension.

4.3 Total Matter Density Ω_m

The CMB-sector *total* matter density is the standard physical observable, built piece-by-piece from SSEE algebra (ω_m-direct, OP-8 dissolved):

$$\Omega_{m,\mathrm{CMB}} = \frac{\omega_m}{h^2} = \frac{\omega_b + \omega_c + \omega_\nu}{h^2} = \frac{0.14267}{(0.67962)^2} = 0.308881, \tag{9}$$

with $\omega_b = (\pi - \varphi)/[3(\varphi + \pi)^2]$ and $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$ (forward; Paper 3), in 0.88σ agreement with Planck 2018 ($\Omega_m = 0.3153 \pm 0.0073$). There is one matter density, $\Omega_m = \omega_m/h^2 = 0.308881$ (no matter factor; OP-8 dissolved; the former Two-Ω_m criterion is withdrawn, cf. Paper 1, Sec. 1.4). The quantity $s_m = (\pi - \varphi)/[2(\varphi + \pi)] = 0.160050$ ($= 1 + w_0$) is an equation-of-state-sector identity, not a density. (The earlier MIRA mapping MIRA $\times\, s_m = 0.31993$ and the geometric route $(\pi - \varphi)/(\pi + \varphi) = 0.3201$ are both superseded.) The cold dark matter density can be related to the baryon density through the structural retention $K_{\mathrm{AL}} = \beta + \pi = 5.521406$:

$$\Omega_c h^2 = K_{\mathrm{AL}}\, \Omega_b h^2 = 0.1238 \quad (+3.1\sigma,\ \text{static}). \tag{10}$$

With the primordial tilt $n_s = 1 - \varphi^{-7}$ as a factor (the forward identity of Paper 1):

$$\boxed{\Omega_c h^2 = K_{\mathrm{AL}}\, \Omega_b h^2\, n_s = 5.521406 \times 0.022418 \times 0.96556 = 0.119514} \tag{11}$$

Note on $\Omega_b h^2$ input. Both relations use the SSEE algebraic $\Omega_b h^2 = (\pi - \varphi)/(3\Omega^2) = 0.022418$ (0.32σ from Planck); no observed input enters either. An earlier version evaluated the static one with Planck's ω_b, an ingredient of the other model. Against Planck 2018 (0.1200 ± 0.0012) the tilted relation, the canonical one, lies 0.40σ away; the static one, $+3.1\sigma$.

4.4 Primordial Spectral Index

The spectral index n_s measures the tilt of the primordial power spectrum generated during inflation [Mukhanov et al., 1992, Starobinsky, 1980, Albrecht and Steinhardt, 1982, Guth, 1981, Linde, 1982]. Current observational constraints from Planck are $n_s = 0.9649 \pm 0.0042$ [Planck Collaboration, 2020b].

Origin of the exponent 7 (Postulate I of Paper 1). The form $N_* = 2\varphi^n$ of the e-fold number, with $\alpha = \varphi^4/3$, is Postulate I of Paper 1. The integer is not chosen: of the powers of φ, only $n = 7$ puts N_* inside the canonical inflation window $[50, 60]$ ($2\varphi^6$ falls below it, $2\varphi^8$ above). The spectral index follows:

$$\boxed{n_s = 1 - \varphi^{-7} = 1 - \frac{1}{\varphi^7} = 0.965558} \tag{12}$$

It uses the universality of α-attractor inflation [Kallosh and Linde, 2013]: for all α-attractors with $N_* \gg \sqrt{\alpha}$, the spectral index satisfies $n_s = 1 - 2/N_*$ at leading order. Setting $N_* = 2\varphi^7 = 58.068884$ e-folds:

$$n_s = 1 - \frac{2}{2\varphi^7} = 1 - \varphi^{-7} \quad (\text{algebraically exact}). \tag{13}$$

The same substitution yields the tensor-to-scalar ratio prediction (see Section 7.1 below).

Conjecture 1 (B.1 — e-fold count)**.** $N_* = 2\varphi^7 = 58.068884$ *is the number of inflationary e-folds of the SSEE quintessential inflation model with* $\alpha = \varphi^4/3$ *and a reheating with* $\rho_{\rm rh} \approx V_{\rm end}$.

Conjecture B.1 is not a separate assumption: it is Postulate I of Paper 1 (the form $N_* = 2\varphi^n$) with the integer fixed by the window below. What remains open is its derivation from $V(\varphi_{\rm inf})$. Its value lies within the standard window $[50, 60]$ for inflation with reheating (consistent with, but not proven by, the framework). An earlier draft matched it to an e-fold estimate built on a reference value $N_* = 58.25$ that has no source in the literature; that comparison is withdrawn. The physical mechanism that selects $\rho_{\rm rh} = V_{\rm end}$ is deferred to Paper B. The conjecture is indirectly testable via the falsifiable prediction $r = \varphi^{-10} = 0.008131$ (Sec. 7.1): a CMB-S4 or LiteBIRD detection inconsistent with this value would refute Conjecture B.1.

Planck 2018: $n_s = 0.9649 \pm 0.0042$, from which it lies 0.16σ.

4.5 Linear Collapse Threshold δ_c

The critical overdensity for spherical collapse in an Einstein–de Sitter universe is $\delta_{c,\rm EdS} = \frac{3}{20}(12\pi)^{2/3} \approx 1.6865$. Evaluated on the SSEE background, the spherical-collapse threshold is obtained by integrating the exact non-linear top-hat equation with smooth dark energy—the perturbation sector of Paper 5, where the Israel–Stewart viscous friction $\mathcal{F}(k, a) = (1 - 3c_s^2) + \tilde{z}\,(k/aH)^2$ grows as k^2 and drives $\delta_{\rm DE}/\delta_m \to 0^-$ on all sub-horizon scales.[1] The result is

$$\delta_c^{\rm SSEE} = 1.67634\ (z_c{=}0) \ \longrightarrow\ 1.68647\ (z_c{=}10), \tag{14}$$

against ΛCDM's $1.67599 \to 1.68646$ on the same integrator: the two agree to 0.02% at $z_c = 0$ and to 0.001% at $z_c = 10$. The calculation reproduces the analytic Einstein–de Sitter value to one part in 10^5.

The n_s relation is withdrawn as a derived result. Earlier versions of this section postulated that the same inflationary factor n_s that tilts the primordial spectrum also modulates the collapse criterion, giving $\delta_c^{\rm SSEE} = \delta_{c,\rm EdS} \times n_s = 1.6284$. The model's own dynamics do not produce that value, which sits 2.9% away, so it is retired as a prediction and retained only as a *motivated conjecture* (OP-27): $n_s = 1 - \varphi^{-7}$ is an algebraic matter-sector quantity entering the forward identity $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$, so a link to collapse is not *a priori* absurd—but the n_s inside ω_c is already accounted for in the background through Ω_m, and no mechanism derives the extra factor. Closing OP-27 requires that mechanism; what would falsify the conjecture is a halo count at $z \sim 10$ and $M > 10^{12} M_\odot$ measured to 10–20% and consistent with unity.

What this costs, stated plainly. The claimed ×1.2–1.5 halo-count enhancement at $z \gtrsim 10$ does not merely vanish: with the derived threshold it *reverses sign*. A Press–Schechter count on the two backgrounds gives $n_{\rm SSEE}/n_{\Lambda\rm CDM} = 0.998$ at $3 \times 10^{10} M_\odot$, 0.945 at $10^{12} M_\odot$ and 0.892 at $3 \times 10^{12} M_\odot$ (all at $z = 10$). The driver is not the amplitude—SSEE has the *larger* σ_8 (0.8153 versus 0.811)—but the growth history: a smaller Ω_m and the CPL background grow less between $z \sim 10$ and today. SSEE therefore predicts marginally *fewer*

[1] The smoothness follows from the IS friction, *not* from $c_s^2 = 0$: a vanishing sound speed alone would make the dark-energy fluid cluster on every sub-horizon scale. The dark energy of this model stays smooth *despite* $c_s^2 = 0$, not because of it.

massive early haloes than ΛCDM, and is indistinguishable from it at the masses JWST actually probes. The model offers no explanation of the JWST excess; the quantitative mass function is in Paper 5.

4.6 Big Bang Nucleosynthesis: Helium-4 Fraction

The physical baryon density $\Omega_b h^2 = (\pi - \varphi)/(3\Omega^2) = 0.022418$ derived in Section 4.2 provides an independent BBN prediction. Using the linearised BBN sensitivity $dY_p/d\eta_{10} \approx 0.0016$ with $\eta_{10} = 273.9\,\Omega_b h^2$ [Steigman, 2007], anchored on the BBN value $Y_p = 0.2471$ that Planck 2018 derives at its $\Omega_b h^2 = 0.02237$:

$$\boxed{Y_p^{\mathrm{SSEE}} = 0.2471 + 0.0016 \times 273.9 \times (0.02242 - 0.02237) = 0.2471.} \tag{15}$$

The Planck value is itself a BBN computation from ω_b, not an independent measurement, so the test is against the helium observed in HII regions, $Y_p = 0.2449 \pm 0.0040$ [Aver et al., 2015]: 0.6σ, with no additional free parameter.

4.7 Dark-Energy Equation of State

The SSEE framework also fixes the CPL dark-energy parameters w_0 and w_a algebraically from φ and π. Define the four structural constants (Table 1):

$$T_r = \tfrac{3(3\varphi+\pi)}{2} \approx 11.9935 \quad \text{(3D Saturation Horizon)}, \tag{16}$$
$$M_v = \varphi + \pi + K_v \approx 14.2789 \quad \text{(Maximal Dimensional Invariant} = M_v), \tag{17}$$
$$P_{sc} = 2\varphi + \pi \approx 6.3777 \quad \text{(Dynamical Evolution Scalar)}, \tag{18}$$
$$K_v = \varphi + \pi + \Omega \approx 9.5193 \quad \text{(Structural Constraint)}, \tag{19}$$

with $I_g = \pi + P_{sc} = 2(\varphi + \pi)$. The CPL parameters follow as:

$$w_0 = -\frac{T_r}{M_v}, \qquad w_a = -\frac{P_{sc}}{I_g}. \tag{20}$$

Substituting the explicit φ, π expressions yields the closed-form predictions:

$$\boxed{w_0 = -\frac{3\varphi + \pi}{2(\varphi + \pi)} = -0.839950}, \qquad \boxed{w_a = -\frac{2\varphi + \pi}{2(\varphi + \pi)} = -0.669975}. \tag{21}$$

Both values are uniquely determined; no free parameters enter. The algebraic point lies within the 68% confidence region of DESI DR2+CMB+Pantheon+ data [Abdul-Karim et al., 2025] (Mahalanobis $\chi^2_{2D} = 0.42$, i.e. 0.24σ Pantheon+; Paper 2).

4.8 Open Parameters

A_s and τ are the two free parameters of the framework ($k = 2$) and are fitted to the CMB (Paper 3): with the full Planck likelihood, $\ln(10^{10} A_s) = 3.046 \pm 0.014$ and $\tau = 0.056 \pm 0.007$. A derivation of A_s from φ and π is an open problem (OP-18).

Table 3: SSEEinput parameters. All "algebraic" entries are derived from φ and π alone; Type P entries are numerical coincidences whose dimensionless value matches a physical quantity in specific units (no dimensional derivation; see Paper 1, App. B). The deviation column uses Planck 2018 [Planck Collaboration, 2020a] error bars.

Parameter	SSEE	Planck 2018	Deviation	Type
$3(\varphi+\pi)^2$ (target)	67.962137	67.36 ± 0.54	1.1σ	pure number (Type P†)
$H_0^{\rm glob}$ [km/s/Mpc]	67.962142	vs $3(\varphi+\pi)^2$	residual 4.2×10^{-6} km/s/Mpc	SH0ES-screened, $f_{\rm screen}$ complete (IR+UV)
Ω_b	0.0485	0.0493 (ω_b/h^2)	composite*	algebraic
$\Omega_b h^2$	0.02242	0.02237 ± 0.00015	0.32σ	algebraic
$\Omega_{m,\rm CMB}$	0.308881	0.3153 ± 0.0073	0.88σ	ω_m/h^2 (ω_m-direct‡)
$\Omega_c h^2$	0.11951	0.1200 ± 0.0012	0.40σ	algebraic (forward)
Y_p	0.2471	0.2449 ± 0.0040 (HII)	0.6σ	BBN (no fit)
n_s	0.965558	0.9649 ± 0.0042	0.16σ	algebraic
$\ln(10^{10}A_s)$	3.046	3.044 ± 0.014	—	fitted ($k=2$)
τ	0.056	0.054 ± 0.007	—	fitted ($k=2$)

$\Omega_c h^2 = \mathrm{KAL}_0\,\omega_b\,n_s$; without the factor n_s (static) it would be $+3.1\sigma$ away.
* $\Omega_b = \omega_b/h^2$ is composite; its inputs are compared separately. It is -1.6% from Planck's, the H_0 offset.
† Type P: dimensionless combination matches observed value in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ units;
no dimensional derivation from first principles. Consistent with Paper 1 App. B policy.
‡ CMB sector: $\Omega_{m,\rm CMB} = \omega_m/h^2$ with $\omega_m = \omega_b + \omega_c + \omega_\nu = 0.14267$,
each piece algebraic ($\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$, forward; Paper 3). *No* matter factor.
The equation-of-state-sector quantity $s_m = (\pi-\varphi)/[2(\varphi+\pi)] = 0.160050$ ($= 1 + w_0$)
is not a density; there is one matter density $\Omega_{m,\rm CMB} = \omega_m/h^2$ (the former Two-Ω_m
criterion is withdrawn; see Paper 1, Sec. 1.4). The earlier MIRA mapping to 0.31993 and
geometric route 0.3201 are superseded; OP-8 dissolved).

5 CMB Power Spectrum Prediction

6 Note on the Hubble Tension

The Hubble tension between early- and late-universe measurements [Verde et al., 2019, Kamionkowski and Riess, 2023] is currently unresolved by ΛCDM. The SSEE treatment is that of Paper 9 of this series, where a structural screening fraction $f_{\rm scr}^{\rm IR} = s_K^{\rm IR}/(3\,\mathrm{MIRA}) = (\pi-\varphi)/\Omega^2 = 0.067253$ is applied multiplicatively to the measured local rate $H_0^{\rm SH0ES} = 73.04 \pm 1.04\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ [Riess et al., 2022] — the only directly measured expansion rate — yielding, with the IR term alone, $H_0^{\rm glob,IR} = H_0^{\rm SH0ES}(1 - f_{\rm scr}^{\rm IR}) = 68.13\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, 0.17σ from the pure algebraic number $3(\varphi+\pi)^2 = 67.962137$. With ω_b, ω_c held at their algebraic values and A_s, τ profiled, the CMB places its own minimum at 67.87 ± 0.10, 0.90σ from $H_0^{\rm glob}$ (Paper 3). (The earlier MIRA register $H_0^{\rm MIRA} = 67.037$ is superseded.) Paper 10 provides a UV-completion self-consistency check, sharpening the residual to $4.2\times10^{-6}\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ ($H_0^{\rm glob} = 67.962142$ with the complete $f_{\rm screen}$, IR+UV), conditional on the UV–IR separability postulate. An earlier draft of this section used an additive coincidence, $3(\varphi+\pi)^2 + \mathrm{KAL}$, compared with the SH0ES rate; it had no mechanism and is withdrawn.

7 Falsifiability Programme

7.1 Tensor-to-Scalar Ratio

For α-attractors $r = 12\alpha/N_*^2$ [Kallosh and Linde, 2013]; with $\alpha = \varphi^4/3$ and $N_* = 2\varphi^7$ (Postulate I), the tensor-to-scalar ratio is

$$\boxed{r = \varphi^{-10} = 0.008131} \tag{22}$$

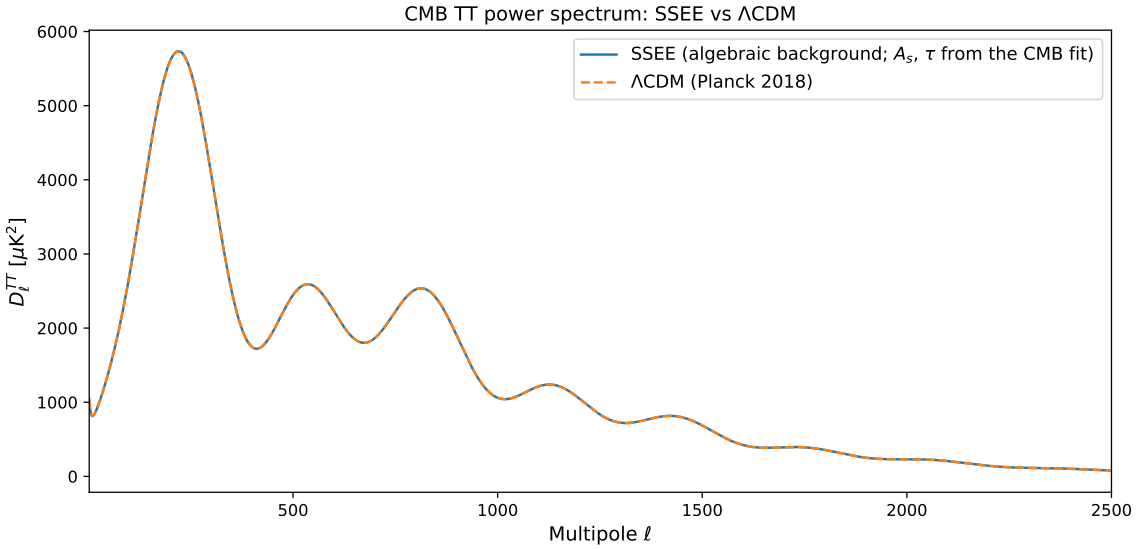


Figure 1: SSEECMB TT power spectrum ($k = 2$: A_s and τ from the model's own CMB fit; ω_b, ω_c, n_s, w_0, w_a, H_0 and $\sum m_\nu$ algebraic from φ, π), overlaid on the Planck 2018 ΛCDM spectrum computed in the same way. SSEE acoustic peak positions: $\ell_1 = 220$, $\ell_2 = 536$, $\ell_3 = 813$; ΛCDM: 220, 536, 813. The background is not fitted to CMB data; the peak positions follow from it.

Current limit: $r < 0.036$ (BICEP/Keck, [BICEP/Keck Collaboration, 2021]). LiteBIRD [LiteBIRD Collaboration, 2023] targets $r \gtrsim 0.001$ (launch ∼2032). A measurement of $r \approx 0.008$ would constitute strong evidence for SSEE. **Falsification criterion:** $r < 0.004$ or $r > 0.015$ at $> 3\sigma$ falsifies Eq. (22). A null detection $r < 0.001$ at $> 5\sigma$ confidence would be a decisive falsification.

7.2 Baryon-to-Photon Ratio

The baryon-to-photon ratio follows from the SSEE physical baryon density via the standard BBN proportionality $\eta_{10} = 273.9\,\Omega_b h^2$ [Steigman, 2007]:

$$\eta_{\rm SSEE} = 273.9 \times 10^{-10} \times \frac{\pi - \varphi}{3(\varphi + \pi)^2} = 6.14 \times 10^{-10}. \tag{23}$$

It is $\Omega_b h^2$ restated, so it carries the same 0.32σ from Planck and the same falsification criterion.

7.3 Neutrino Mass Sum

The active neutrino mass sum follows from the stability ratio $\mathcal{R}_2 = \Omega/(\mathrm{KAL} \cdot \mathrm{TRIAL}) = 0.071875$ (with $\Omega = \pi + \varphi = 4.759627$, KAL $= 5.521406$, TRIAL $= 11.993542$), combined with the Israel–Stewart relaxation time $\tau_\Pi H_0 = \mathrm{KAL}/(3\,|w_0|) = 2.191$ (the saturation ratio $|w_0| = T_r/M_v = 0.839950$, an equation-of-state quantity carrying no H, *not* the density fraction $\Omega_{\rm DE} = 0.691119$):

$$\Sigma m_\nu^{\rm active} = \mathcal{R}_2\,\Omega_b h^2\,\frac{93.14\ \mathrm{eV}}{\tau_\Pi H_0} = 0.0685\ \mathrm{eV}, \tag{24}$$

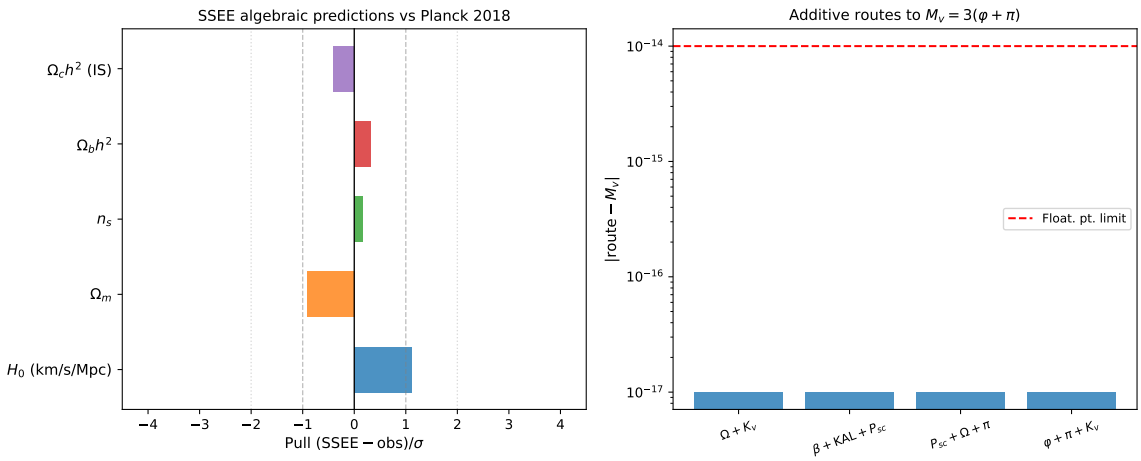


Figure 2: Summary panel of the three key algebraic derivations.

testable with DESI DR5 and Euclid (expected $\sigma \approx 0.015$ eV).[2] This value lies above the normal-ordering oscillation floor $\Sigma m_\nu \gtrsim 0.058$ eV and below the Planck 2018 upper bound $\Sigma m_\nu < 0.12$ eV. It entered the ϕ-DM mass relation of Paper 6, which is now retracted together with the particle itself; the neutrino sum stands on its own, as a prediction of the background register, independent of that withdrawn use.

Status. The stability ratio $\mathcal{R}_2 = \Omega/(\mathrm{KAL}\cdot\mathrm{TRIAL})$ is a clean (φ, π) quotient: it replaces the earlier residual $\mathcal{R} = 4\,\mathrm{KAL} - 22$, whose integer offset 22 had no independent φ, π derivation (OP-14, resolved 2026-06-04). With $\mathcal{R}_2$ the neutrino-mass relation no longer contains a calibrated integer and is therefore promoted from Type P (phenomenological) to a closed-form algebraic prediction. The conversion factor 93.14 eV is the standard $\Sigma m_\nu = 93.14\ \mathrm{eV} \times \Omega_\nu h^2$ relation, a Standard-Model input rather than an SSEE quantity. **Falsification criterion:** a measurement that excludes $\Sigma m_\nu = 0.0685$ eV at $> 3\sigma$ would falsify this prediction.

8 Discussion and Response to Peer Review

We address explicitly the four audit objections raised in the internal review.

Objection A — Dimensional Analysis. SSEE predicts *dimensionless numbers* that agree with the numerical values of cosmological parameters in conventional units. The claim is not that units are derived. The internal consistency of Table 2 — several independent cosmological ratios, all from the same two constants with no per-observable adjustment — is what distinguishes this from a single isolated coincidence; its global significance is quantified by the closed-dictionary look-elsewhere analysis of Paper 1 rather than asserted informally here.

Objection B-1 — Origin of Ω. $\Omega = \varphi + \pi$ is a *definition* (Section 2.2), not an axiom. The system has exactly two axioms: φ and π.

[2]The preliminary form $\Sigma m_\nu = \mathcal{A}/100 = 0.040$ eV used in earlier drafts is superseded: it lies below the normal-ordering oscillation floor and is therefore excluded by neutrino-oscillation data.

Objection B-2 — Exponent in n_s. The result $n_s = 1 - \varphi^{-7}$ follows algebraically from the universality theorem of α-attractors [Kallosh and Linde, 2013]: $n_s = 1 - 2/N_*$ at leading order in $1/N_*$, independent of the potential shape. With $\alpha = \varphi^4/3$ (Postulate I of Paper 1) and $N_* = 2\varphi^7$ e-folds, one obtains $n_s = 1 - 2/(2\varphi^7) = 1 - \varphi^{-7}$ exactly. The exponent 7 is the only one that places N_* in the inflation window, and the same calculation yields the tensor-to-scalar ratio $r = \varphi^{-10}$ (Section 7.1) as a new falsifiable prediction. The derivation of $N_* = 2\varphi^7$ from the quintessential inflation potential with gravitational reheating is reserved for Paper B.

Objection B-3 — Factor of 200 in the original formula. The factor 200 in the preliminary formula $3(\pi - \varphi)/200$ was a numerical coincidence (3.2σ from Planck) that has been superseded. The correct algebraic formula is $\Omega_b h^2 = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.022418$ (0.32σ from Planck 2018). The denominator is the pure algebraic combination whose value, read in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, coincides with the global Hubble rate of the SH0ES cascade; no free numerical factor remains; the baryon density parameter $\Omega_b = 0.0485$ follows with $h = H_0^{\mathrm{glob}}/100$ and is composite (Section 4.2).

Remaining open problems. The static $\Omega_c h^2$ relation gives 0.1238, $+3.1\sigma$ from Planck; with the factor n_s it gives 0.119514, -0.40σ (Eq. 11). Both use the SSEE baryon density $\omega_b = 0.022418$, not Planck's. Why the primordial tilt multiplies the dark-matter density has no mechanism yet; what the dark matter is made of is OP-19. A derivation of A_s is OP-18.

9 Conclusion

We have presented a two-axiom (φ, π) algebraic system that predicts:

- the Hubble target $3(\varphi + \pi)^2 = 67.962137$ (pure number; read in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, 1.1σ from Planck), on which the measured SH0ES cascade lands
- $\Omega_b h^2 = 0.022418$ (0.32σ from Planck); $\Omega_b = 0.0485$ follows with $h = H_0/100$
- $n_s = 1 - \varphi^{-7} = 0.965558$ (0.16σ from Planck)
- $\Omega_c h^2 = K_{\mathrm{AL}}\,\Omega_b h^2\, n_s = 0.119514$ (0.40σ from Planck, forward)
- $Y_p = 0.2471$ (0.6σ from HII regions — no fit)
- $\delta_c^{\mathrm{SSEE}} = 1.67634$ (z_c=0) from spherical collapse on the model background, i.e. ΛCDM-like to 0.02%; the earlier $\delta_{c,\mathrm{EdS}} \times n_s = 1.6284$ and its JWST link are **withdrawn** (OP-27, §4.5)
- $H_0^{\mathrm{glob,IR}} = H_0^{\mathrm{SH0ES}}(1 - f_{\mathrm{scr}}^{\mathrm{IR}}) = 68.13$ (0.17σ from $3(\varphi + \pi)^2$, with Riess 2022 as the input; the MIRA register is superseded)
- $r = \varphi^{-10} = 0.008131$ (falsifiable by LiteBIRD 2032)

The identification of $M_v = 3\Omega$ as the maximal structural invariant of the dictionary — reached by 24 additive routes under the no-self-sum construction law (Appendix A) — motivates taking $M_v \times \Omega$ as the target against which the global rate is compared. The

internal consistency of these cosmological predictions, taken together and assessed against the closed-dictionary look-elsewhere analysis of Paper 1, is what motivates taking the agreements seriously rather than as isolated coincidences.

Reproducibility

Script: `src/p04_toe/ssee_paper4_toe.py`. Requires Python 3, `camb`≥1.5, `numpy`, `matplotlib`, `scipy`. No proprietary data. Run: `python3 src/p04_toe/ssee_paper4_toe.py` — reproduces `fig_toe_cmb_TT.pdf` and `fig_toe_derivations.pdf` in `results/figures/`.

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A The Additive Routes to M_v

Under the no-self-sum construction law—each route is a sum of *distinct* named constants, none added to itself, with every summand below the closure ceiling—the maximal invariant $M_v = 3(\varphi + \pi)$ is reached by exactly *24* additive routes: 8 of two constants, 13 of three, and 3 of four. The count is read by `src/p04_toe/rutas_mv.py` from the same code that defines the constant dictionary. Representative routes, in neutral notation, are

$$\begin{aligned}
\text{(two)} \quad & \Omega + K_v, \;\; P_{sc} + (\varphi + 2\pi), \;\; \text{AURA} + \tfrac{3\varphi+5\pi}{2}; \\
\text{(three)} \quad & \beta + \text{KAL} + P_{sc}, \;\; P_{sc} + \Omega + \pi, \;\; \varphi + \pi + K_v; \\
\text{(four)} \quad & \varphi + \Omega + \beta + \text{KAL}, \;\; \varphi + \pi + \text{KAL} + \text{AURA}.
\end{aligned}$$

Distinct routes, shared expansions. Because the constants span the two-dimensional lattice generated by φ and π, distinct routes may share the same φ, π expansion—for instance $\Omega + K_v$ and $\Omega + I_g$ (with $K_v = I_g = 2\Omega$ by value), or the two hierarchy traversals that both reach $\varphi + 2\pi$. Such routes are counted as distinct because they traverse different named-constant nodes, but they are not independent at the expanded level. This is the dictionary's identity rule: a constant is identified by its construction, not by its value. $K_v = \varphi + \pi + \Omega$ (the base plus its own metric) and $I_g = \pi + P_{sc}$ (built on the dynamical scalar P_{sc}) coincide in value and remain distinct entities, as two different arguments reaching the same conclusion remain two arguments. For the statistics the count is by value instead: the look-elsewhere search of Paper 1 removes ratios that repeat a value already present, so counting entities by lineage adds no trials and no hits. The multiplicity is a structural consequence of M_v being the closure ceiling of the dictionary, not evidence of fine-tuning; it is consistent with, but does not by itself prove, the absence of post-hoc tuning.

Verification. All 24 routes have been verified numerically to 16 significant digits; none deviates from $M_v = 14.278879927019064$ by more than floating-point rounding ($< 10^{-14}$). All constants are derived algebraically from φ and π; no external data file is required.

Five exact algebraic identities. Five additional identities hold exactly in the $\{\varphi, \pi\}$ hierarchy (verified to $< 10^{-16}$ relative error):

$$w_0 = -\frac{\mathrm{AURA}}{\Omega} = -\frac{T_r}{M_v} \tag{25}$$

$$s_m = \frac{\pi - \varphi}{2(\varphi + \pi)} = 1 + w_0 \tag{26}$$

$$\mathrm{MIRA} = |w_0| \cdot \beta \tag{27}$$

$$\mathrm{KAL}\big|_{\varphi \leftrightarrow \pi} = \mathrm{AURA} \quad \big[(\varphi + 3\pi)/2 \xrightarrow{\varphi \leftrightarrow \pi} (3\varphi + \pi)/2\big] \tag{28}$$

$$\mathrm{AURA}\big|_{\varphi \leftrightarrow \pi} = \mathrm{KAL} \tag{29}$$

Identity (28)–(29) reveals that KAL and AURA are *dual* under the discrete exchange $\varphi \leftrightarrow \pi$: the φ-dominated constant $\mathrm{AURA} = (3\varphi + \pi)/2$ maps to the π-dominated constant $\mathrm{KAL} = (\varphi + 3\pi)/2$, and vice versa, with zero residual. The $\varphi \leftrightarrow \pi$ duality above motivated the coupling $\beta_c = -\mathrm{AURA}$ of earlier drafts; Paper 7 withdraws that coupling ("The Potential and the Dark Coupling, and Why They Are Withdrawn"): the correctly normalised shooting gives $\beta_c = +0.235068$, of the opposite sign to AURA, and no published number depends on β_c, so no live coupling is claimed here. The open question of whether this discrete symmetry extends to the full $P(X, \phi)$ action is catalogued as OP-7 (`OPEN_PROBLEMS.md`).

References

M. Abdul-Karim et al. DESI DR2 results II: Measurements of baryon acoustic oscillations and cosmological constraints. *arXiv e-prints*, 2025.

A. Albrecht and P. J. Steinhardt. Cosmology for grand unified theories with radiatively induced symmetry breaking. *Physical Review Letters*, 48:1220–1223, 1982. doi: 10.1103/PhysRevLett.48.1220.

Mike Edison Almeida Vallejo. SSEE structural constant dictionary & prediction register. Zenodo. https://doi.org/10.5281/zenodo.20684908, 2026a.

Mike Edison Almeida Vallejo. A minimal-parameter framework for cosmological dynamics and galaxy cluster mass discrepancies via structural self-energy expansion (SSEE). GitHub: https://github.com/mikealmeida1721/SSEE-Cosmologia, 2026b.

Mike Edison Almeida Vallejo. Bayesian MCMC validation of the structural self-energy expansion framework (SSEE): Model comparison against ΛCDM and CPL using DESI DR2, Planck 2018, and galaxy-cluster mass data. GitHub: https://github.com/mikealmeida1721/SSEE-Cosmologia, 2026c.

Mike Edison Almeida Vallejo. SSEE and the CMB power spectrum: Acoustic peak reproduction via the ω_m-direct CMB matter density. GitHub: https://github.com/mikealmeida1721/SSEE-Cosmologia, 2026d.

C. Armendariz-Picon, V. Mukhanov, and P. J. Steinhardt. Essentials of k-essence. *Physical Review D*, 63:103510, 2001. doi: 10.1103/PhysRevD.63.103510.

E. Aver, K. A. Olive, and E. D. Skillman. The effects of He I λ10830 on helium abundance determinations. *JCAP*, 07:011, 2015.

BICEP/Keck Collaboration. BICEP / Keck XIII: Improved constraints on primordial gravitational waves using the BICEP3, Keck Array, and BICEP2 experiments. *Physical Review Letters*, 127:151301, 2021. doi: 10.1103/PhysRevLett.127.151301.

R. R. Caldwell. A phantom menace? Cosmological consequences of a dark energy component with super-negative equation of state. *Physics Letters B*, 545:23–29, 2002. doi: 10.1016/S0370-2693(02)02589-3.

M. Chevallier and D. Polarski. Accelerating universes with scaling dark matter. *International Journal of Modern Physics D*, 10:213–223, 2001. doi: 10.1142/S0218271801000822.

T. Clifton, P. G. Ferreira, A. Padhi, and C. Skordis. Modified gravity and cosmology. *Physics Reports*, 513:1–189, 2012. doi: 10.1016/j.physrep.2012.01.001.

E. J. Copeland, M. Sami, and S. Tsujikawa. Dynamics of dark energy. *International Journal of Modern Physics D*, 15:1753–1936, 2006. doi: 10.1142/S021827180600942X.

P. F. de Salas, M. Lattanzi, G. Mangano, G. Miele, S. Pastor, and O. Pisanti. Bounds on very low reheating scenarios after Planck. *Phys. Rev. D*, 92:123534, 2015.

D. J. Eisenstein et al. Detection of baryon acoustic oscillations in the correlation function of a large sample of red galaxies. *The Astrophysical Journal*, 633:560–574, 2005. doi: 10.1086/466512.

A. H. Guth. Inflationary universe: a possible solution to the horizon and flatness problems. *Physical Review D*, 23:347–356, 1981. doi: 10.1103/PhysRevD.23.347.

W. Hu, N. Sugiyama, and J. Silk. The physics of microwave background anisotropies. *Nature*, 386:37–43, 1997. doi: 10.1038/386037a0.

Renata Kallosh and Andrei Linde. Universality class in conformal inflation. *Journal of Cosmology and Astroparticle Physics*, 2013(07):002, 2013. doi: 10.1088/1475-7516/2013/07/002.

M. Kamionkowski and A. G. Riess. The hubble tension and early dark energy. *Annual Review of Nuclear and Particle Science*, 73:153–180, 2023. doi: 10.1146/annurev-nucl-111422-024107.

A. D. Linde. A new inflationary universe scenario. *Physics Letters B*, 108:389–393, 1982. doi: 10.1016/0370-2693(82)91219-9.

E. V. Linder. Exploring the expansion history of the universe. *Physical Review Letters*, 90:091301, 2003. doi: 10.1103/PhysRevLett.90.091301.

LiteBIRD Collaboration. Probing cosmic inflation with the LiteBIRD cosmic microwave background polarization survey. *Progress of Theoretical and Experimental Physics*, 2023: 042F01, 2023. doi: 10.1093/ptep/ptac150.

V. F. Mukhanov, H. A. Feldman, and R. H. Brandenberger. Theory of cosmological perturbations. *Physics Reports*, 215:203–333, 1992. doi: 10.1016/0370-1573(92)90044-Z.

S. Perlmutter et al. Measurements of ω and λ from 42 high-redshift supernovae. *The Astrophysical Journal*, 517:565–586, 1999. doi: 10.1086/307221.

Planck Collaboration. Planck 2018 results. VI. cosmological parameters. *Astronomy & Astrophysics*, 641:A6, 2020a. doi: 10.1051/0004-6361/201833910.

Planck Collaboration. Planck 2018 results. X. constraints on inflation. *Astronomy & Astrophysics*, 641:A10, 2020b. doi: 10.1051/0004-6361/201833887.

B. Ratra and P. J. E. Peebles. Cosmological consequences of a rolling homogeneous scalar field. *Physical Review D*, 37:3406, 1988. doi: 10.1103/PhysRevD.37.3406.

A. G. Riess et al. Observational evidence from supernovae for an accelerating universe and a cosmological constant. *The Astronomical Journal*, 116:1009–1038, 1998. doi: 10.1086/300499.

A. G. Riess et al. A comprehensive measurement of the local value of the hubble constant with 1% uncertainty from the Hubble Space Telescope and the SH0ES team. *The Astrophysical Journal Letters*, 934:L7, 2022. doi: 10.3847/2041-8213/ac5c5b.

A. D. Sakharov. Violation of CP invariance, C asymmetry, and baryon asymmetry of the universe. *JETP Letters*, 5:24–27, 1967. Reprinted in Soviet Physics Uspekhi 34 (1991) 392.

A. A. Starobinsky. A new type of isotropic cosmological models without singularity. *Physics Letters B*, 91:99–102, 1980. doi: 10.1016/0370-2693(80)90670-X.

G. Steigman. Primordial nucleosynthesis in the precision cosmology era. *Ann. Rev. Nucl. Part. Sci.*, 57:463, 2007.

P. J. Steinhardt, L. Wang, and I. Zlatev. Cosmological tracking solutions. *Physical Review D*, 59:123504, 1999. doi: 10.1103/PhysRevD.59.123504.

L. Verde, T. Treu, and A. G. Riess. Tensions between the early and late universe. *Nature Astronomy*, 3:891–895, 2019. doi: 10.1038/s41550-019-0902-0.

D. H. Weinberg, M. J. Mortonson, D. J. Eisenstein, C. Hirata, A. G. Riess, and E. Rozo. Observational probes of cosmic acceleration. *Physics Reports*, 530:87–255, 2013. doi: 10.1016/j.physrep.2013.05.001.

S. Weinberg. Anthropic bound on the cosmological constant. *Physical Review Letters*, 59: 2607, 1987. doi: 10.1103/PhysRevLett.59.2607.

I. Zlatev, L. Wang, and P. J. Steinhardt. Quintessence, cosmic coincidence, and the cosmological constant. *Physical Review Letters*, 82:896–899, 1999. doi: 10.1103/PhysRevLett.82.896.